

SE ZG651 / SS ZG653: Software Architectures
Situated Learning Assignment
Computer Science and Information Systems
Work Integrated Learning Programmes Division
Birla Institute of Technology and Science, Pilani
Section A: Assignment Brief

- **Course:** SE ZG651 / SS ZG653 -- Software Architectures
- **Instructor:** HARVINDER SINGH JABBAL
- **Semester:** S2-25
- **Organisation:** [Ernst & Young / Jio Platforms / Verizon / Lloyd / Siemens]
- **Component:** EC-1
- **Assignment:** Situated Learning (Assignment #1)
- **Weightage:** 5 Marks
- **Submission:** PDF (PPT format exported to PDF)

A.1 Learning Objectives

- Apply software architecture basics and architectural tactics in a real-world workplace context.
- Analyse and design software architecture diagrams for domain-specific systems.

A.2 Core Concept

- Software Architecture Basics, Architectural Tactics, Architecturally Significant Requirements (ASRs), and Architectural Views/Diagrams.

A.3 Scope and Constraints

- The system identified must be an existing system from the student's workplace.
- Submissions must follow the specified PPT structure and be exported as a PDF file.

A.4 Apply Tasks

Using an existing software system from your workplace as context, perform the following tasks:

1. **System Context & Requirements:** Define the purpose/goal (1 slide) and key functional & non-functional requirements (2 slides).
2. **ASRs & Tactics:** Build a Utility Tree for Architecturally Significant Requirements (1 slide) and document the tactics used to achieve the top 5 ASRs (1 slide).
3. **Architecture Diagrams:** Draw 4 architecture diagrams: Context Diagram, Module Decomposition, Component & Connection Diagram, and Deployment Diagram (4 slides).
4. **Operation & Reflection:** Describe how the system works (1 slide) and detail key learnings (1 slide).

A.5 Evaluation Rubric

Component	Description	Marks
Anchor	Clearly identifies an existing system from your workplace with sufficient context	1
Apply	Correctly applies architecture concepts via Utility Tree, top 5 tactics, and 4 architectural diagrams	2
Analyse	Evaluates system requirements and describes how components interact to solve the domain problem	1
Reflect	Articulates key learnings and operational limitations within the workplace context	1
Total		5

Section B: Student Submission

B.1 Submission Form

- **Name:** _____
- **ID No:** _____
- **Industry / Domain:** _____
- **Organisation Type:** _____
- **Instructor prompt:** Select an existing workplace system to document its goal, requirements, ASR utility tree, architectural tactics, and 4 core architectural diagrams.

B.2 Anchor (1 mark)

- Identify an existing system from your workplace.
- **Slide 1:** Purpose and primary goal of the system.

B.3 Apply (2 marks)

- **Slides 2–3:** Key functional and non-functional requirements.
- **Slide 4:** Utility tree of Architecturally Significant Requirements (ASRs).
- **Slide 5:** Tactics used to achieve the top 5 ASRs.
- **Slides 6–9:** Software Architecture Diagrams:
 - Context diagram
 - Module decomposition
 - Component & Connection diagram
 - Deployment diagram

B.4 Analyse (1 mark)

- **Slide 10:** Detailed description of how the system works based on your architectural models.
- Analyze gaps between theoretical design tactics and actual operational needs in your workplace.

B.5 Reflect (1 mark)

- **Slide 11:** Key learnings gained from analysing this architecture.
- Identify constraints (e.g., scale, legacy tech stack, security) that influence future architectural improvements.